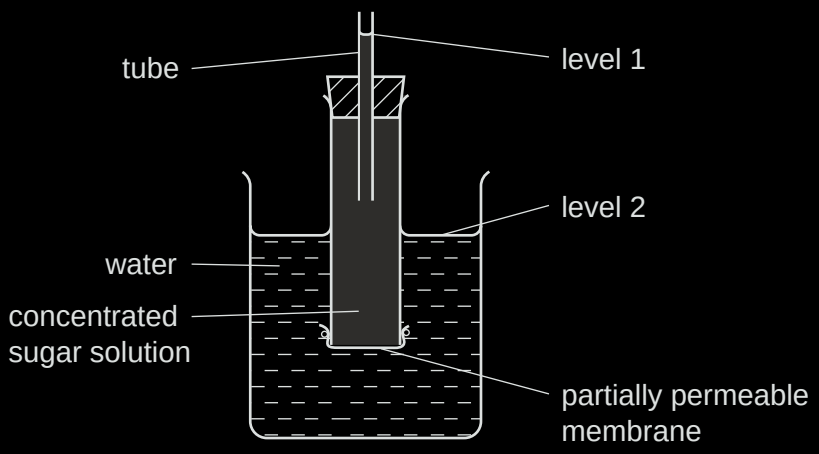
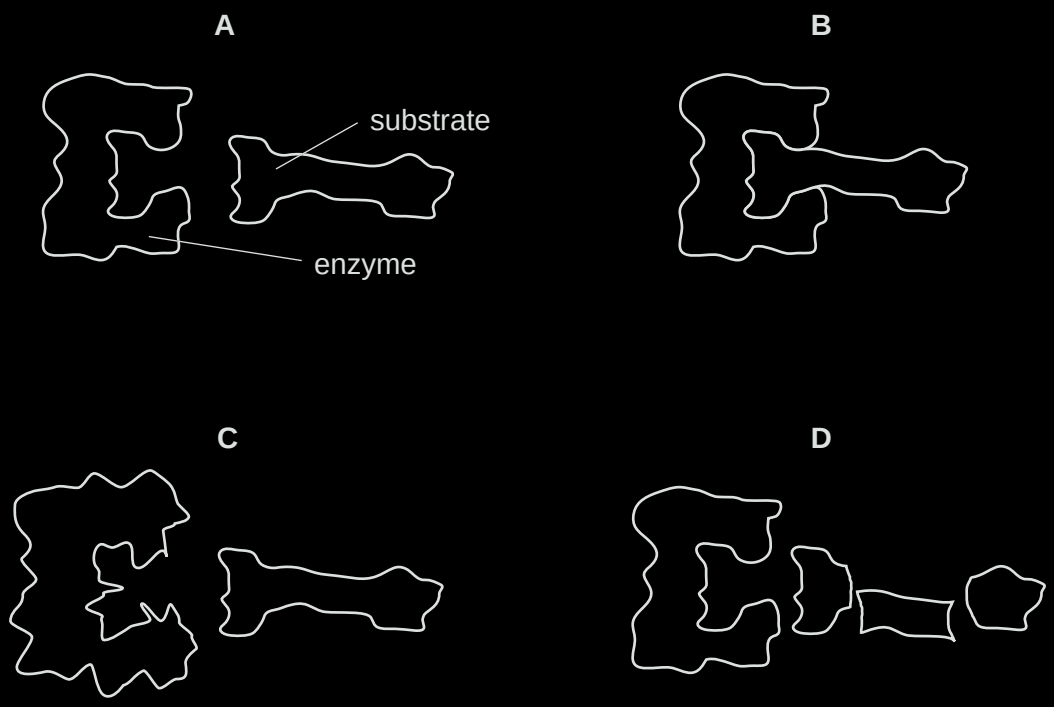
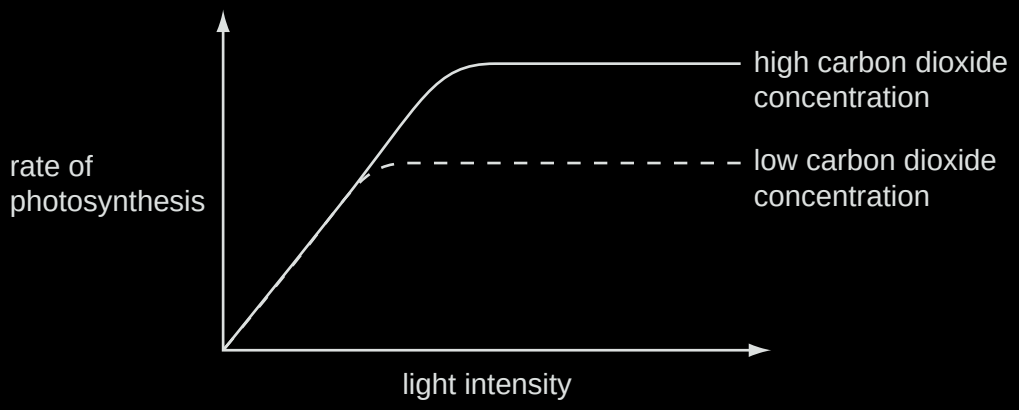


Which molecules will move across the partially permeable membrane and which changes in levels will occur?

	molecules	level 1	level 2
A	sugar	fall	rise
B	water	fall	rise
C	sugar	rise	fall
D	water	rise	fall



5 The graph shows the effect of changing light intensity on the rate of photosynthesis in a plant at two different carbon dioxide concentrations.



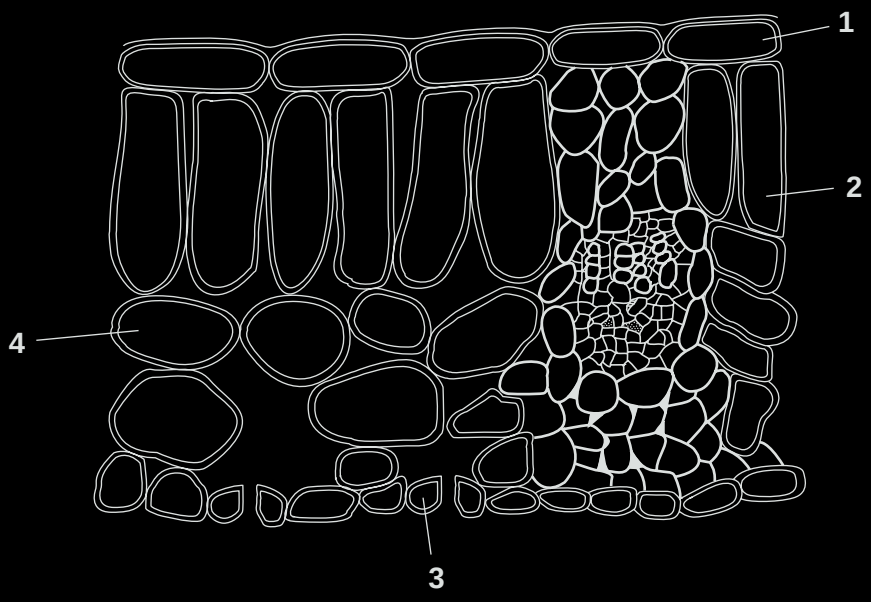
Which statement is correct?

A At low light intensities carbon dioxide is the limiting factor.

B At high light intensities carbon dioxide is the limiting factor.

C When the carbon dioxide concentration is high, there is no limiting factor.

D When carbon dioxide concentration is low, the plant cannot photosynthesise.



Which cells would contain chloroplasts?

A **1** and **2** only

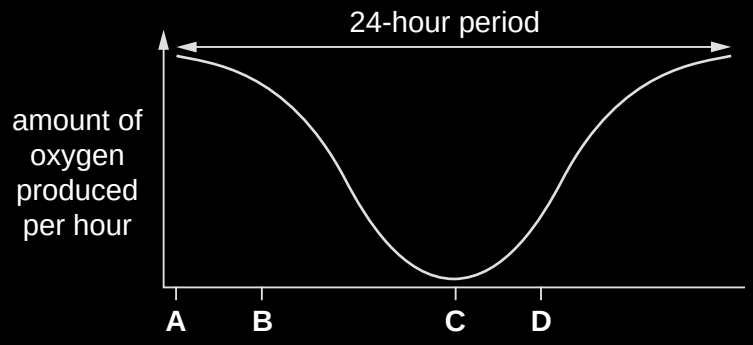
B **2** only

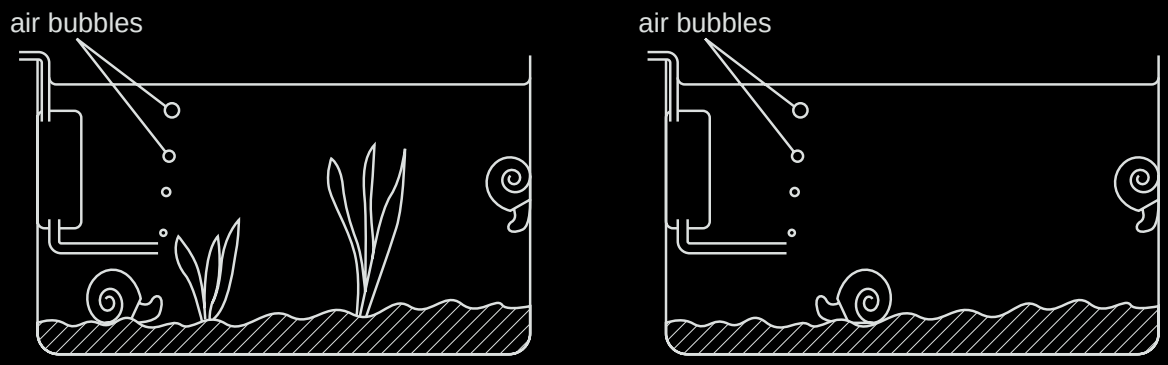
C **2** and **3** only

D **2**, **3** and **4** only

7 The graph shows the amount of oxygen produced by a green plant, growing outdoors, during a 24-hour period.

Which letter represents midday?





tank **X**
contains animals and plants

tank **Y**
contains animals

After a week, all the animals in tank **Y** show signs of distress.

This was because the animals have run out of

A carbon dioxide.

B food.

C nitrate.

D oxygen.

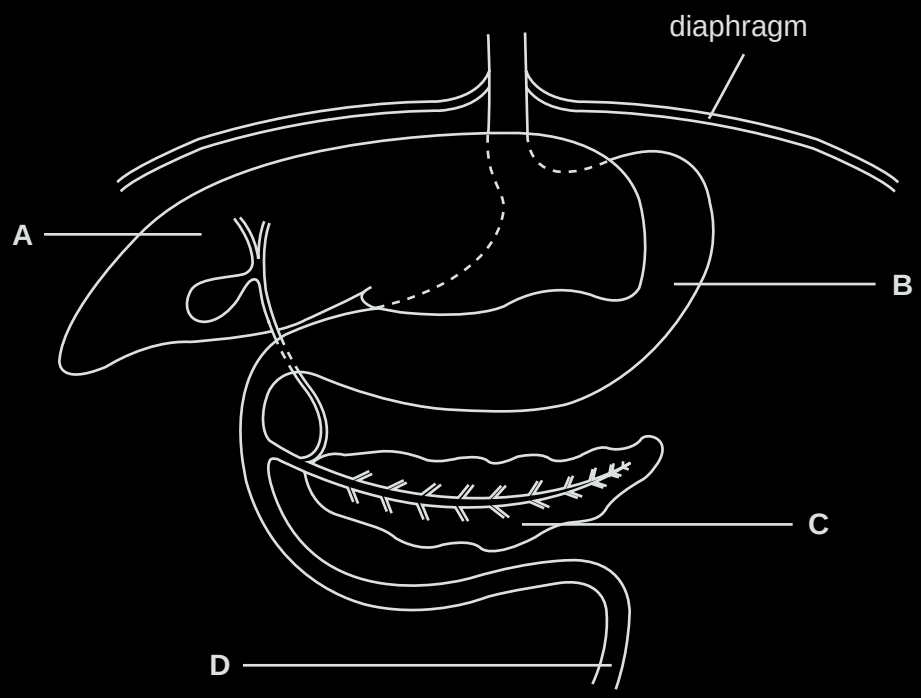
9 Which substances are needed in the diet to prevent rickets?

A calcium and vitamin C

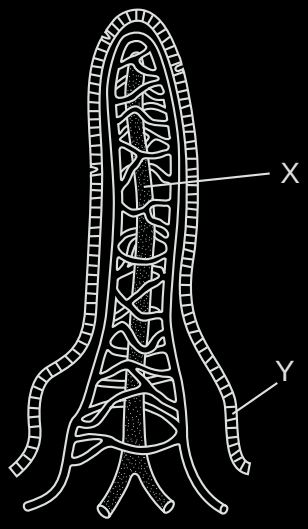
B calcium and vitamin D

C iron and vitamin C

D iron and vitamin D

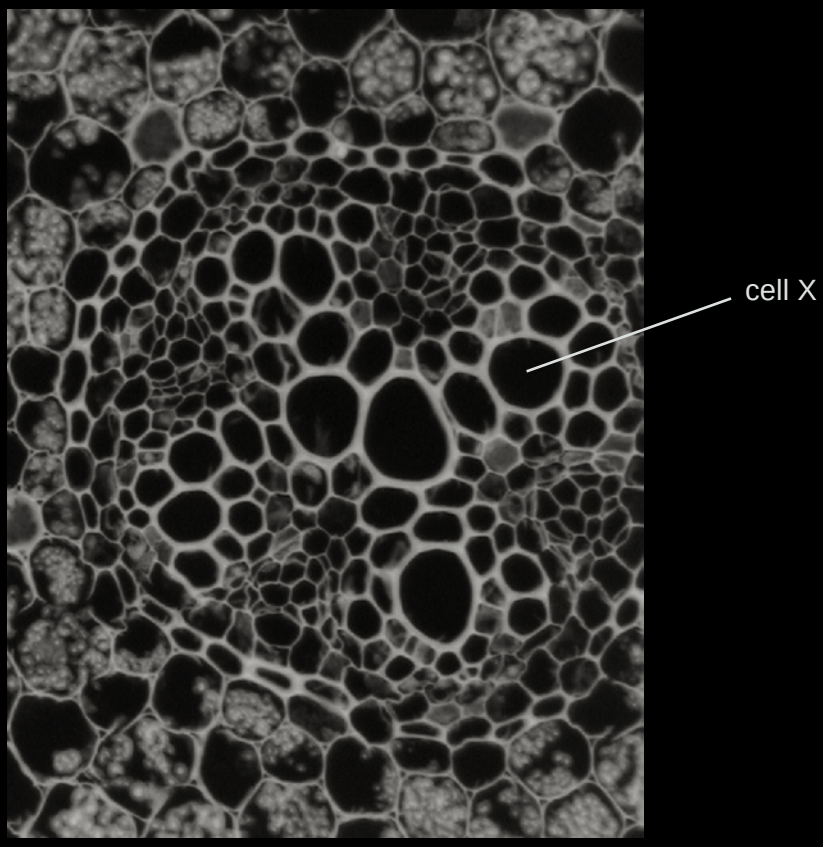


11 The diagram shows a section through a villus.



What is a function of structure X and of structure Y?

	X	Y
A	to absorb amino acids	to digest starch
B	to carry blood	to secrete mucus
C	to transport fats	to secrete enzymes
D	to transport glucose	to help peristalsis



Samples of the contents of cell X were tested.

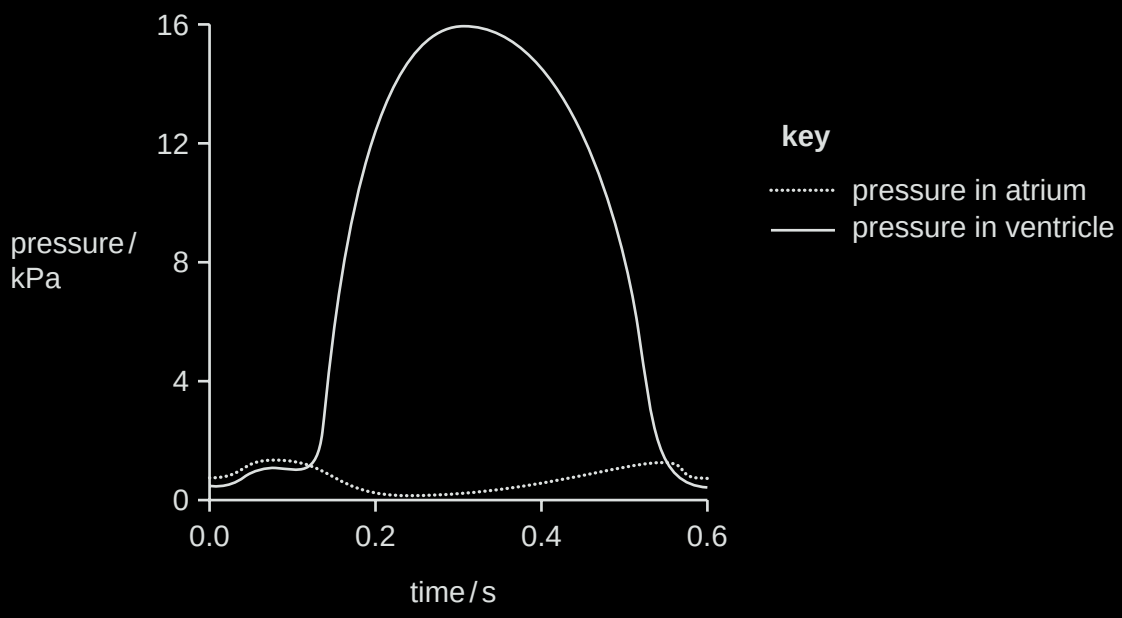
What results are expected?

	Benedict's reagent	iodine
A	+	+
B	+	–
C	–	+
D	–	–

key

+ = positive result

– = negative result



What is the number of times that this person's heat beats in one minute, while at rest?

A 60 **B** 70 **C** 100 **D** 120

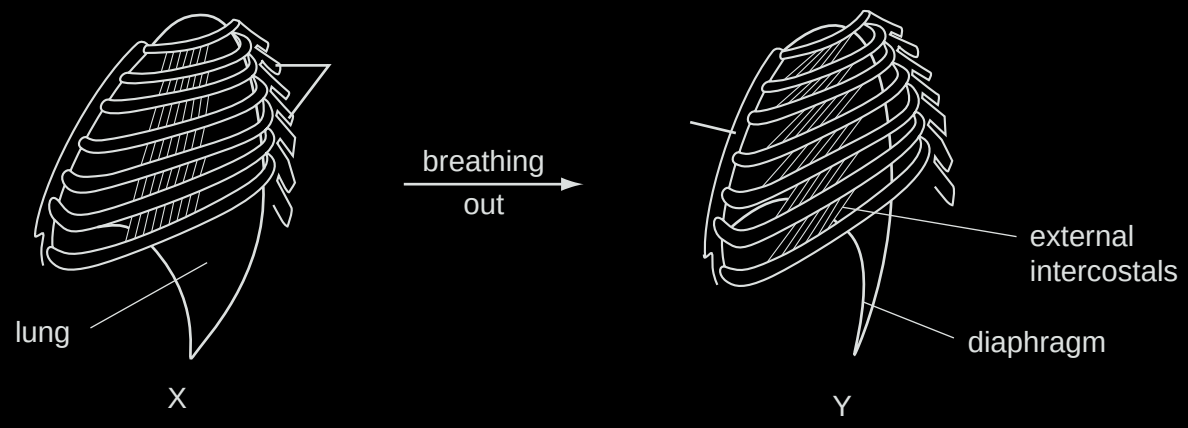
16 Which statement explains why humans are said to have a double circulation?

A As blood circulates it passes twice through the heart.

B Each side of the heart has two chambers.

C Each side of the heart has two valves.

D There are two different sets of arteries leaving the heart.



Which muscles relax in moving from position X to position Y?

	diaphragm	external intercostals
A	no	no
B	no	yes
C	yes	no
D	yes	yes

18 Which substance is produced in both aerobic respiration in humans and anaerobic respiration in yeast?

A carbon dioxide

B ethanol

C lactic acid

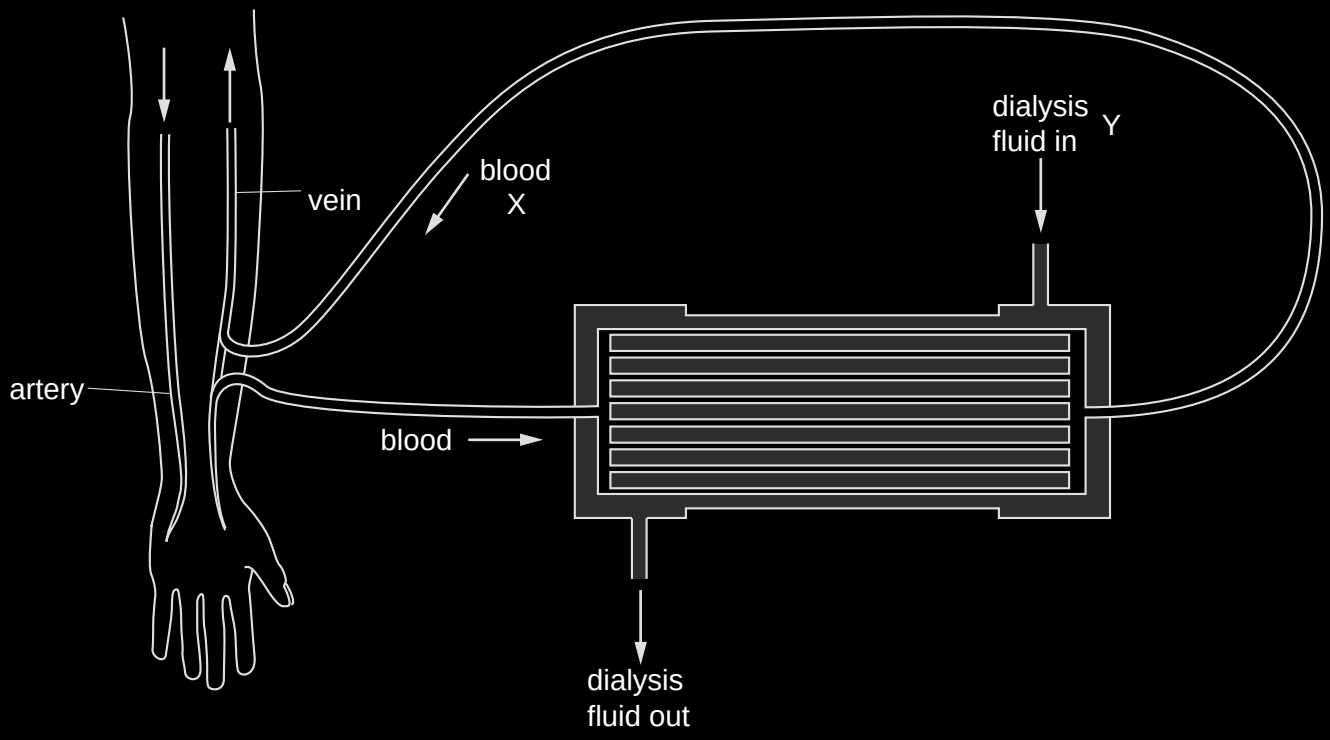
D water

19 The table shows ventilation rates of an adult while resting and while exercising.

adult	volume of air inhaled per breath / cm^3	number of breaths per minute	volume of air exchanged per minute / cm^3
resting	400	20	8000
exercising	1200	40	?

What is the volume of air exchanged per minute while exercising?

A 16 000 cm^3 **B** 24 000 cm^3 **C** 32 000 cm^3 **D** 48 000 cm^3



Which substances have the lowest concentration at X and the highest concentration at Y?

	lowest at X	highest at Y
A	glucose	salts
B	salts	glucose
C	urea	water
D	water	urea

21 The body can regulate both its temperature and the amount of water in its cells.

What are these processes?

A assimilation

B excretion

C homeostasis

D osmosis



	1	2	3
A	cerebellum	cerebrum	pituitary gland
B	cerebrum	cerebellum	pituitary gland
C	pituitary gland	cerebrum	cerebellum
D	pituitary gland	cerebellum	cerebrum

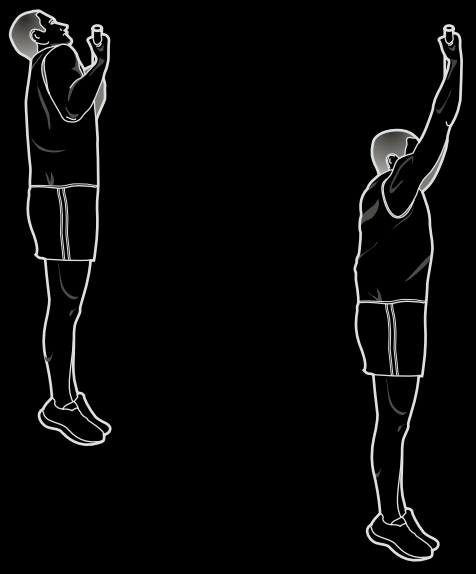
24 What structures cover the pupil of a human eye?

A conjunctiva and cornea

B conjunctiva and sclera

C cornea and retina

D retina and sclera



P Q

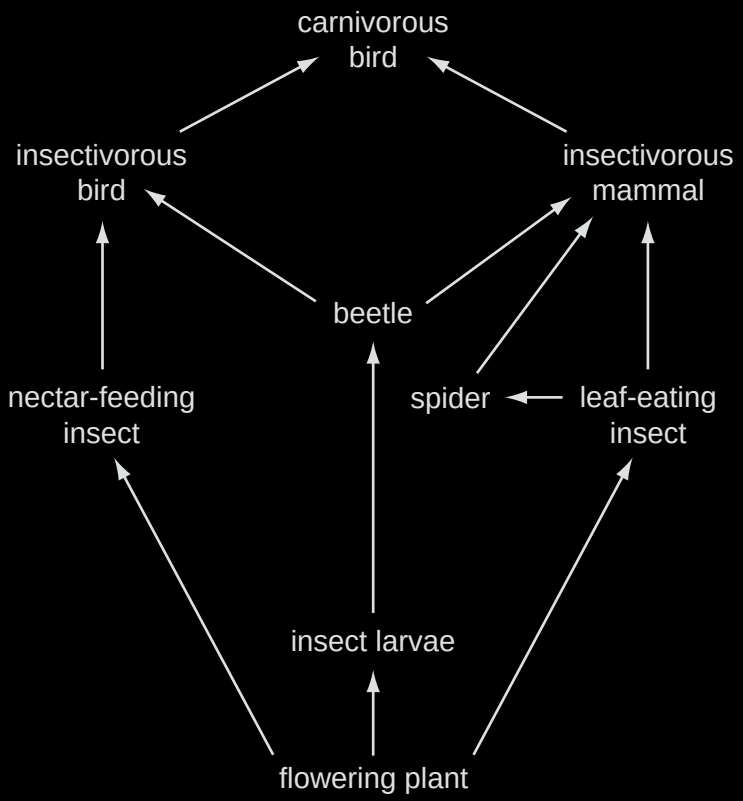
In moving from the upper position P to the lower position Q, which muscles and joint movements occur?

	biceps	triceps	ball and socket joint
A	contracts	contracts	flexes
B	contracts	relaxes	rotates
C	relaxes	contracts	extends
D	relaxes	relaxes	rotates

26 Tar and carbon monoxide are present in tobacco smoke.

What are their effects on health?

	tar	carbon monoxide
A	causes high blood pressure	damages haemoglobin
B	causes high blood pressure	is addictive
C	causes lung cancer	damages haemoglobin
D	causes lung cancer	is addictive



Which is a primary consumer?

A beetle

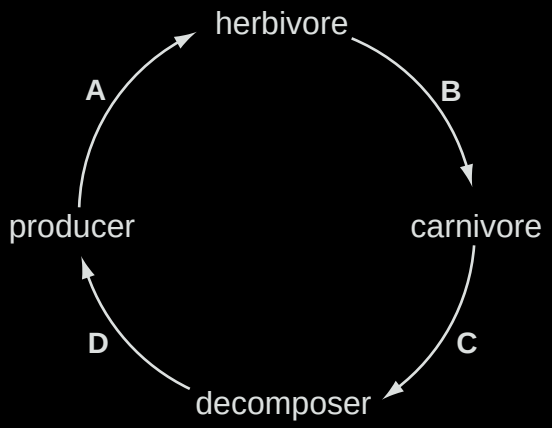
B carnivorous bird

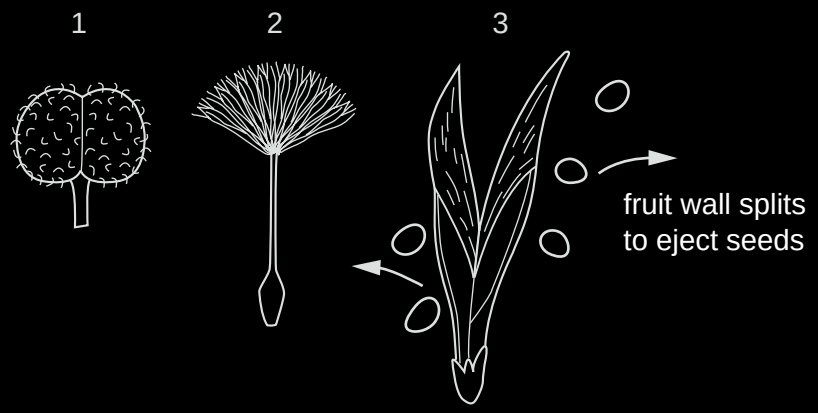
C insectivorous mammal

D nectar-feeding insect

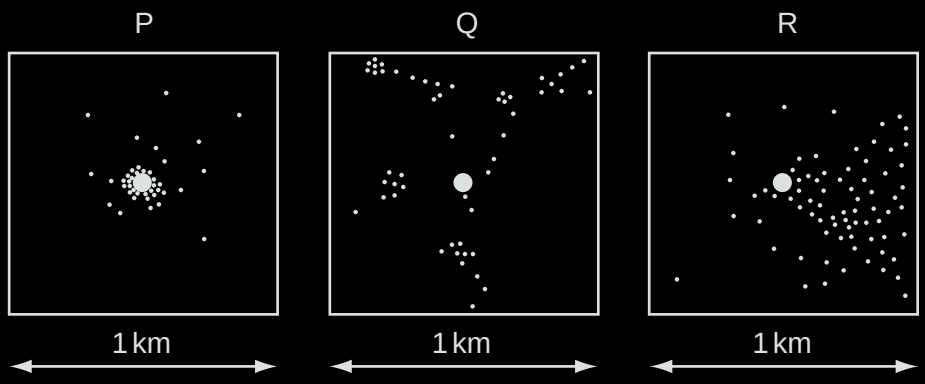
31 The diagram shows how **carbon** circulates in nature.

Through which stage does most **energy** flow?





P, Q and R are maps showing where the parent plants and their seedlings are growing.

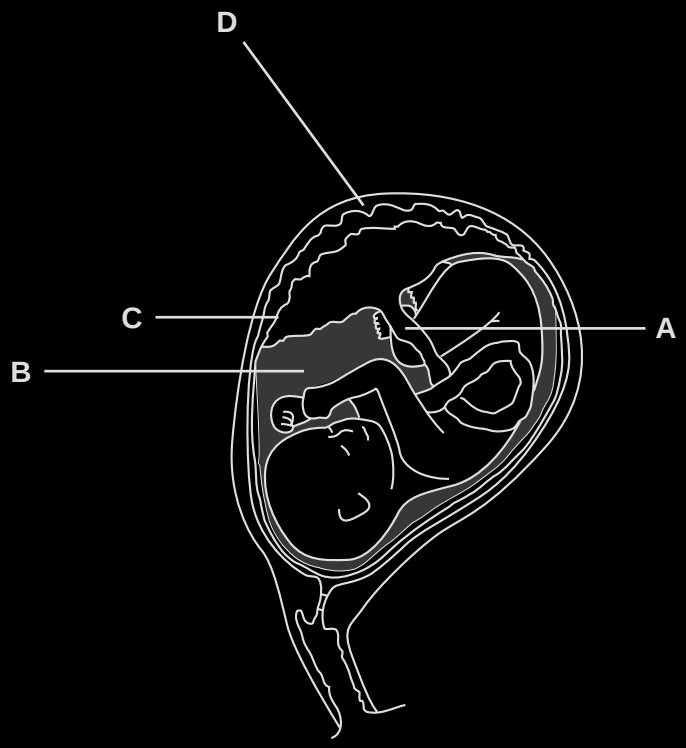


Which map relates to which species?

	species 1	species 2	species 3
A	P	Q	R
B	P	R	Q
C	Q	R	P
D	R	Q	P

36 What is true for syphilis?

	cause	symptoms develop after	treatment
A	bacterium	7-10 days	antibiotics
B	bacterium	number of years	vaccine
C	virus	7-10 days	antibiotics
D	virus	number of years	vaccine



38 Dillip and Shabnam made four statements about themselves.

	Dillip	Shabnam
1	I am a boy.	I am a girl.
2	I am 150 cm tall.	I am 153 cm tall.
3	I am not very good at games.	I am good at games.
4	My blood group is A.	My blood group is AB.

Which statements describe characteristics that show discontinuous variation?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4